

Synthesis Review: Taylor Polynomials and Remainder Bounds

Answer Key

AP Calculus BC

Quick Answers

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| 1. the third derivative = $8e^{2x}$, the estimate of e to the 0.2 = 1.22133 | 6. the error bound = 0.0000217, — |
| 2. the estimate = 0.36843 | 7. the second derivative = $-\frac{1}{4x^{\frac{3}{2}}}$, the estimate of the square root of 4.4 = 2.0975 |
| 3. the Lagrange bound = 0.11326, — | 8. the exact sum = e^{-1} , the error bound = $\frac{1}{24}$ |
| 4. the third derivative = $\frac{2}{x^3}$, the estimate of $\ln(1.2)$ = 0.18267 | 9. the Lagrange bound = 0.015625, — |
| 5. the estimate = 0.8944 | 10. the third derivative = $(x+3)e^x$, the estimate = 0.4035, — |
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What is verified

15 of 19 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 3, 6, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus BC

LIM-8 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 19 tagged checks.

1. $f(x) = e^{2x}$, $P_3(x) = 1 + 2x + 2x^2 + \frac{4}{3}x^3$. (a) Find f''' . (b) Estimate $f(0.1)$.

Solution (a): Each differentiation brings down a factor of 2, so after three:

$$f'''(x) = 8e^{2x}$$

That matches the cubic coefficient: $f'''(0)/3! = 8/6 = \frac{4}{3}$.

Solution (b): $P_3(0.1) = 1 + 0.2 + 0.02 + \frac{4}{3}(0.001)$.

$$P_3(0.1) \approx 1.22133$$

2. $x \cos x \approx x - \frac{x^3}{2} + \frac{x^5}{24}$. Estimate the value at $x = 0.4$.

Solution: Multiplying the cosine series by x shifts every exponent up by one, giving those three terms. At $x = 0.4$:

$$0.4 - \frac{0.064}{2} + \frac{0.01024}{24} = 0.4 - 0.032 + 0.00042667.$$

So the estimate is

$$0.36843$$

3. Lagrange bound for P_3 of e^x at $x = 1$ on $[0, 1]$. (a) The bound. (b) Why $M = e$?

Solution (a): $|R_3(1)| \leq \frac{M|1-0|^4}{4!} = \frac{e}{24}$. $\frac{e}{24} \approx 0.11326$

Solution (b) — instructor-judged. M must bound $|f^{(4)}|$ over the whole interval. Every derivative of e^x is e^x again, and e^x is increasing, so on $[0, 1]$ its largest value is $e^1 = e$. Full credit justifies the choice rather than quoting a number.

4. $f(x) = \ln x$ about $a = 1$. (a) Find f''' . (b) Estimate $\ln(1.2)$ with P_3 .

Solution (a): $f'(x) = x^{-1}$, $f''(x) = -x^{-2}$, $f'''(x) = 2x^{-3}$. $f'''(x) = \frac{2}{x^3}$

At $a = 1$ this is 2, and $2/3! = \frac{1}{3}$, the cubic coefficient shown.

Solution (b): With $x - 1 = 0.2$: $0.2 - \frac{0.04}{2} + \frac{0.008}{3}$. $P_3(1.2) \approx 0.18267$

5. $\frac{1}{1+3x^2} \approx 1 - 3x^2 + 9x^4$. Estimate the value at $x = 0.2$.

Solution: Substituting $u = -3x^2$ into the geometric series gives those terms. At $x = 0.2$:

$$1 - 3(0.04) + 9(0.0016) = 1 - 0.12 + 0.0144.$$

So the estimate is

0.8944

(The true value is $1/1.12 \approx 0.892857$; the next term, $-27x^6$, accounts for most of the gap.)

6. $\cos(0.5)$ estimated through the x^4 term. (a) Alternating error bound. (b) What did you check?

Solution (a): The first omitted term is the x^6 term, of size $\frac{0.5^6}{6!} = \frac{0.015625}{720}$. ≈ 0.0000217

Solution (b) — instructor-judged. The alternating bound requires two things: the series alternates in sign at this input, and the term sizes $\frac{0.5^{2k}}{(2k)!}$ decrease to zero. Both hold for the cosine series at $x = 0.5$, so the remainder is at most the first omitted term. Full credit checks both conditions before quoting the bound.

7. $f(x) = \sqrt{x}$ about $a = 4$. (a) Find f'' . (b) Estimate $\sqrt{4.4}$ with P_2 .

Solution (a): $f'(x) = \frac{1}{2}x^{-1/2}$ and $f''(x) = -\frac{1}{4}x^{-3/2}$, i.e. $f''(x) = -\frac{1}{4x\sqrt{x}}$

At $a = 4$ this is $-\frac{1}{32}$, and $-\frac{1}{32}/2! = -\frac{1}{64}$, the quadratic coefficient shown.

Solution (b): With $x - 4 = 0.4$: $2 + \frac{0.4}{4} - \frac{0.16}{64} = 2 + 0.1 - 0.0025$. $P_2(4.4) = 2.0975$

8. $\sum_{n \geq 0} (-1)^n / n!$. (a) Exact sum and decimal. (b) Error bound after four terms.

Solution (a): This is the Maclaurin series for e^x at $x = -1$.

$$e^{-1} \approx 0.3679$$

Solution (b): The terms alternate and their sizes $\frac{1}{n!}$ decrease to zero, so the error after the terms $n = 0, 1, 2, 3$ is at most the size of the $n = 4$ term, $\frac{1}{4!}$.

$$\frac{1}{24} \approx 0.0417$$

9. Lagrange bound for P_3 of $\ln(1+x)$ at $x = 0.5$, with $f^{(4)}(x) = -6/(1+x)^4$. (a) The bound. (b) Choosing M .

Solution (a): $|R_3(0.5)| \leq \frac{M(0.5)^4}{4!}$ with $M = 6$, so the bound is $\frac{6(0.0625)}{24}$. 0.015625

Solution (b) — instructor-judged. $|f^{(4)}(x)| = \frac{6}{(1+x)^4}$ is a positive constant over a growing denominator, so it is largest where $1+x$ is smallest — at the left end, $x = 0$, where it equals 6. Taking $M = 6$ therefore bounds the fourth derivative across the whole interval. Full credit explains why the maximum sits at that end rather than asserting the value.

10. $f(x) = xe^x$, $P_3(x) = x + x^2 + \frac{x^3}{2}$. (a) Find f''' . (b) Estimate $f(0.3)$. (c) Which bound?

Solution (a): Differentiating xe^x repeatedly gives $(x+1)e^x$, then $(x+2)e^x$, then

$$f'''(x) = (x+3)e^x$$

At 0 this is 3, and $3/3! = \frac{1}{2}$, the cubic coefficient shown.

Solution (b): $P_3(0.3) = 0.3 + 0.09 + \frac{0.027}{2}$.

$$P_3(0.3) = 0.4035$$

Solution (c) — instructor-judged. Use the Lagrange form. The Maclaurin series for xe^x is $x + x^2 + \frac{x^3}{2} + \frac{x^4}{6} + \dots$, all of whose terms are positive at a positive input, so the series does not alternate and the alternating remainder bound simply does not apply. The Lagrange form has no such condition: bound $|f^{(4)}|$ on $[0, 0.3]$ and divide by $4!$. Full credit names the alternation condition as the thing that decides between the two.